

Frozen pyrochlore ice and the $\mathbb{Z}_2$ curl: a reduction of the periodicity conjecture, with proofs in one dimension and two and certificates in three

Lyndon Drake, with Claude Code (Anthropic Claude 5 models)

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Abstract

A state of pyrochlore spin ice is frozen when no hexagon of the diamond lattice can be flipped. Every frozen zero-flux state of the cubic cells of side two and three is invariant under a nonzero translation, and the conjecture is that this holds at every side. This note reduces the conjecture to a parity statement. The curl of a state on a hexagon is the parity of the number of its six bonds oriented from one sublattice to the other; a circulating hexagon has odd curl, so a state with even curl everywhere is frozen. Four statements are proved in full: a missing dipole axis forces a half-box translation symmetry (Lemma A); every hexagon of a frozen state has exactly two reversals of its flow, by a count over the whole torus (Lemma L); on a cell one cube thick, where the model is square ice with dominoes as its moves, a frozen state with a single kink has even curl everywhere and such states are classified by their diagonals (Theorem 2D); and a state that is a sum modulo two of three one-dimensional profiles, one per axis, has a one-cube period whenever it obeys the ice rule (the profile lemma). Two statements carry machine-checked certificates rather than proofs: that a frozen state with all six dipole directions has even curl (Theorem H, DRAT proofs checked by drat-trim on five cells), and that an ice state with even curl has a one-cube period (Theorem P, DRAT proofs on six cells including the cubic cell of side four, together with linear algebra over $\mathbb{F}_2$ and solver checks showing that every even-curl ice state on those cells is a sum of profiles). Together H and P give the conjecture on every cell whose extents are all at least two, and describe every frozen state with six directions as three one-dimensional profiles. What remains is a proof of H in three dimensions and of the profile description in general.

1 The model and the language of dipoles

The diamond lattice has vertices on two sublattices, A and B, with $B = A + (1, 1, 1)$ in units of a quarter of the conventional cubic edge, and four bonds at each vertex along

$d_0 = (1, 1, 1)$, $d_1 = (1, -1, -1)$, $d_2 = (-1, 1, -1)$, $d_3 = (-1, -1, 1)$. A state assigns an arrow to every bond, and the ice rule requires two arrows in and two out at every vertex. On a periodic cell of $a \times b \times c$ conventional cubes there are $8abc$ vertices, $16abc$ bonds and $16abc$ hexagons, the hexagons being the closed six-bond walks of zero displacement (on a small torus the graph has other six-cycles, which wind around the cell and are excluded). Flipping a hexagon reverses its six arrows; the move preserves the ice rule exactly when the arrows circulate, and it preserves the flux, the net arrow count through each coordinate plane. A state is **frozen** when no hexagon circulates.

A vertex with two arrows in and two out has a net moment along one of $\pm e_x, \pm e_y, \pm e_z$: with the out-bonds at an A vertex (in-bonds at a B vertex) forming the pair $\{i, j\}$, the moment is proportional to $d_i + d_j$, so $\{0, 1\}$ and $\{2, 3\}$ give $\pm e_x$, $\{0, 2\}$ and $\{1, 3\}$ give $\pm e_y$, and $\{0, 3\}$ and $\{1, 2\}$ give $\pm e_z$. The axis of the moment is the **type** of the vertex. An ice state is exactly a field of moments $n(v)$ in which every bond of direction d satisfies $[d \cdot n(A) > 0] = [d \cdot n(B) > 0]$ at its two ends.

The bonds form chains along the six $\langle 110 \rangle$ directions: a chain of the family $\{i, j\}$ alternates bonds of directions d_i and d_j and steps by $d_i - d_j$, so the families $\{0, 1\}$ and $\{2, 3\}$ lie in the (100) planes, $\{0, 2\}$ and $\{1, 3\}$ in (010), and $\{0, 3\}$ and $\{1, 2\}$ in (001); call them the x -, y - and z -chains. A vertex is a **kink** of a chain when both of the chain's bonds at it point in or both point out, and a pass-through otherwise. A vertex is a kink of exactly the two chains perpendicular to its type and a pass-through of the other four. Along a chain the flow keeps its sense between kinks and reverses at each, and kinks alternate between sources and sinks. Every chain is closed on the torus.

A hexagon omits one bond direction and uses one chain of each of the three axes twice, at antipodal pairs of its vertices. It circulates exactly when every one of its six vertices is a pass-through for the chain the hexagon uses there, with a consistent sense. In terms of moments, the circulating pattern on the hexagon $A_1 B_1 A_2 B_2 A_3 B_3$ that omits d_3 is $n(A_1) \in \{+x, +z\}$, $n(B_1) \in \{+y, +z\}$, $n(A_2) \in \{-x, +y\}$, $n(B_2) \in \{-x, -z\}$, $n(A_3) \in \{-y, -z\}$, $n(B_3) \in \{+x, -y\}$, or the negative of that.

2 **Lemma A: a missing axis forces a half-box symmetry**

Lemma A. Let S be an ice state on the cell (a, b, c) , with box $B = (4a, 4b, 4c)$, in which no vertex has type x . Then S is invariant under every translation in

$$\Lambda_x = (\mathbb{Z}(0, 2, 2) + B\mathbb{Z}^3) \cap (\mathbb{Z}(0, 2, -2) + B\mathbb{Z}^3).$$

For a cubic cell of side L that lattice modulo the box is $\{0, (0, 2L, 2L)\}$, the half-box shift in the two coordinates other than x . The same holds for y and z .

Proof. A vertex is a kink of a $\{0, 1\}$ chain iff its type is x . With no such vertex every $\{0, 1\}$ chain is kink-free, so its sense is constant along it, so every d_0 bond on it carries the same arrow and so does every d_1 bond. Consecutive d_0 bonds of a chain are related by the step $(0, 2, 2)$, and every d_0 or d_1 bond lies on such a chain, so the arrows of all d_0 and d_1 bonds are invariant under $(0, 2, 2)$. The $\{2, 3\}$ chains make the arrows of all d_2 and d_3 bonds invariant under $(0, 2, -2)$. A translation that is a multiple of $(0, 2, 2)$ modulo the box and a multiple of $(0, 2, -2)$ modulo the box fixes every bond. For the cubic box, $k(0, 2, 2) \equiv m(0, 2, -2) \pmod{2L}$ needs $k \equiv m$ and $k \equiv -m \pmod{2L}$, so $k \equiv 0 \pmod{L}$. $\square$

Neither freezing nor zero flux is used. Zero flux is the statement that on each sublattice every axis carries as many $+$ as $-$ moments (the kinks of a closed chain alternate source and sink, and a $+x$ kink is a source at an A vertex and a sink at a B vertex), so a zero-flux state in which every axis occurs has all six directions. That is how the six-direction hypothesis of Theorem H enters: with Lemma A it turns the periodicity conjecture into a statement free of flux.

3 **The curl**

For a state $s \in \mathbb{F}_2^E$, with $s_e = 1$ when the arrow on bond e points from A to B, define the **curl** on a hexagon h as $\kappa_h(s) = \sum_{e \in h} s_e \pmod{2}$. Walking a hexagon in cycle order, consecutive vertices lie on opposite sublattices, so a circulating hexagon has its bits alternating and exactly three of them set.

Lemma C. A state with even curl on every hexagon is frozen. $\square$

The converse fails: the states of Lemma A with a missing axis are frozen with odd

hexagons, and so are frozen states in nonzero flux sectors. Theorem H below says the converse holds once all six directions are present.

Orient a hexagon cyclically and read each of its bonds as forward or backward along the cycle. A vertex of the hexagon is a **reversal** when the two hexagon bonds at it point both in or both out, that is, when it is a kink of the chain the hexagon uses there. In bits, a vertex is a reversal iff its two hexagon bonds carry equal bits, again because consecutive vertices alternate sublattice. The number of reversals is even, and zero reversals is circulation. The three B vertices contribute an odd number of bit changes between the bit and the forward/backward reading, so **a hexagon has odd curl iff its number of forward bonds is even**.

4 Lemma L: exactly two reversals

Lemma L. In a frozen state every hexagon has exactly two reversals.

Proof. Twelve hexagons pass through a vertex v , two for each of the six pairs of bonds at v . The vertex is a reversal for a hexagon iff the hexagon's pair at v is one of the two chains kinked at v , so v is a reversal for exactly four hexagons. Summing over vertices, the reversals number $4|V| = 32abc = 2|H|$. A frozen state has no circulating hexagon, so every hexagon has at least two reversals, hence exactly two. $\square$

The count is an average over the torus, and the lemma is not a local one: on the $(4, 4, 4)$ cell a hexagon with four reversals is consistent with freezing on the 610 hexagons within three steps of it, and only the whole torus excludes it. With exactly two reversals the two runs of the flow have lengths $(1, 5)$, $(2, 4)$ or $(3, 3)$, and by the parity count **an odd hexagon has its two reversals at distance two, an even one has them adjacent or antipodal**. Every hexagon of a frozen state therefore has one source and one sink, with both arcs directed from the one to the other, and every vertex is the source of two hexagons (those through its out-chain) and the sink of two.

The chain form of the curl. Fix a two-step path v_k, v_{k+1}, v_{k+2} along an E -chain, with bond directions i and j . Two hexagons contain it, one for each of the two remaining directions n ; the one with parameter n uses the chain $\{i, n\}$ at v_k and $\{j, n\}$ at v_{k+2} , and the axes of those two chains are the two axes other than E , in one order for one choice of n and the other order for the other. In a frozen state that hexagon is odd iff its two

reversals are v_k and v_{k+2} , by Lemma L, iff v_k has the axis of $\{i, n\}$ and v_{k+2} the axis of $\{j, n\}$. Hence:

Proposition. A frozen state has even curl everywhere iff along every chain any two pass-through vertices at distance two have the same type.

Both directions were also checked by solver on $(2, 2, 2)$, $(2, 3, 3)$ and $(3, 3, 3)$. So Theorem H, stated below, says that in a frozen state with all six directions the types of the pass-through vertices along a chain change only across a kink of that chain.

5 One cube thick: square ice with domino moves

Project along x and write $u = (y + z)/2$, $w = (y - z)/2$. The vertices of the cell (a, b, c) lie on $4a$ grids $x = 0, \dots, 4a - 1$, grid x being one parity class of (u, w) on the torus $T = \mathbb{Z}^2 / \langle (2b, 2b), (2c, -2c) \rangle$. Each bond joins grids x and $x + 1$ and projects to a unit step. The bonds between grids x and $x + 1$, the **slab** x , are unit u -steps on the lines $w \equiv x/2 \pmod{2}$ when x is even and unit w -steps on the lines $u \equiv (x + 1)/2$ when x is odd, and these lines are exactly the x -chains. Every hexagon is a 2×1 **domino**: two consecutive steps on each of two adjacent parallel lines, in slabs $x - 1$ and $x + 1$, closed by one step on each of two consecutive lines of slab x . The count is $4bc$ dominoes per middle slab, $16abc$ in all.

On a thin cell, $a = 1$, every line occurs once, so **the thin cell is square ice on T whose flip moves are the dominoes rather than the plaquettes**; a unit plaquette is a non-contractible four-cycle around x and flipping it changes the flux. The six vertex types of square ice are the six moments: with σ_u, σ_w the senses of the row and column through a pass-through, $+y = (+, +)$, $-y = (-, -)$, $+z = (+, -)$, $-z = (-, +)$; the type- x vertices are the c -vertices, where the row sense and the column sense both flip. In six-vertex language type y is an a -vertex, type z a b -vertex.

For a unit square Q with edge senses β (bottom), ρ (right), τ (top), λ (left), write $P(Q) = \beta\rho\tau\lambda$. The product of the senses around a domino is $P(Q_L)P(Q_R)$ (the shared rung cancels), a circulating domino has product $\sigma^3(-\sigma)^3 = -1$, and $P(Q) = y(BL)y(TR)$ where $y(v) = r(v)t(v)$ is the product of the right and upper senses at v , which equals $+1$ exactly at the a -vertices. So $P \equiv +1$ says the a -vertices form complete NE diagonals of T , and equally that the b -vertices form complete SE diagonals.

Theorem 2D. (i) If every unit square has $P = +1$ the state is frozen. (ii) If the state is frozen and has a c -vertex, every unit square has $P = +1$.

Proof of (i). The senses around a domino multiply to $+1$ and a circulating domino has product -1 . $\square$

Proof of (ii). At a c -vertex the right and upper senses satisfy $rt = -1$, the left and lower being $-r$ and $-t$. Consider two adjacent rows $w, w + 1$ carrying opposite constant senses $\sigma, -\sigma$ on a run of edges I , with rungs t_u the column edges between them.

(AP) The domino $[u_0, u_0 + 2] \times [w, w + 1] \subset I$ circulates iff $t_{u_0} = -\sigma$ and $t_{u_0+2} = \sigma$, for either orientation, so along I at each parity of u the rungs form a word $\sigma^*(-\sigma)^*$.

(END) If I is maximal, its ends p (left) and q (right) are c -vertices of row w or row $w + 1$, and $t_p = -\sigma, t_q = \sigma$ in all four cases: a c of row w at p has $r = \sigma$ hence $t = -\sigma$; a c of row $w + 1$ at p has $r = -\sigma$, hence $t = \sigma$ and the rung $d = -t = -\sigma$; symmetrically at q .

(ALT) By (AP) and (END) all rungs at the parity of p equal $-\sigma$ and all at the parity of q equal σ ; so $q - p$ is odd and adjacent rungs alternate along I . Since on I one has $P(Q_{u,w}) = \sigma(-\sigma)t_u t_{u+1} = -t_u t_{u+1}$, every unit square of an anti-parallel run with an endpoint is good. On a parallel run (both rows σ), $P(Q_{u,w}) = t_u t_{u+1}$. The same three facts hold for columns, with rungs the row edges.

Let Q be a bad square with no c -corner. Rows and columns pass through all four corners, so $P = [\beta = \tau][\lambda = \rho]$ in ± 1 , and exactly one of the row pair and the column pair is anti-parallel at Q . If the rows are, (ALT) shows their run has no endpoint, so rows w and $w + 1$ are c -free constant lines. By (AP) applied cyclically the rungs are constant at each parity, and the bad square makes the two parities equal: every column has sense ρ on the edge $(w, w + 1)$, hence, passing through the c -free rows, on $[w - 1, w + 2]$. The rungs between rows $w + 1$ and $w + 2$ are then all ρ . A run of row $w + 2$ with sense σ bounded by two c -vertices of row $w + 2$ is an anti-parallel run whose endpoint rungs are σ and $-\sigma$ by (END), and both equal ρ , which is impossible. So row $w + 2$ has no such run and is c -free and constant, the same follows for its rungs with row $w + 3$, and inductively every row is c -free, against the hypothesis. The column case is identical.

Let Q be bad with a c -corner. By the half-turn symmetry and $P = y(BL)y(TR)$ we may take $BL = (u, w)$ a c -vertex and $TR = (u + 1, w + 1)$ an a -vertex (if TR is also c the square is good). Write β, λ for the right and upper senses at BL , so $\lambda = -\beta$, and

ρ, τ for the lower and left senses at TR , so $\rho = \tau$. If BR and TL are not c : when $\beta = \rho$ the columns u (sense $-\beta$ above w) and $u + 1$ (sense β) are anti-parallel on a run starting at the c -vertex BL with $w + 1$ in its interior, and (ALT) gives $r_{w+1}(u + \frac{1}{2}) = -\beta$, but that edge is $\tau = \beta$; when $\beta = -\rho$ the rows w (sense β right of BL) and $w + 1$ (sense $-\beta$) are anti-parallel on a run starting at BL with $u + 1$ interior, and (ALT) gives $t_{u+1} = \beta$, but $t_{u+1} = \rho = -\beta$. If BR is a c -vertex and TL is not, the a -condition at TR and the b -condition forced at TL give $\tau = \rho = \beta$, the columns $u, u + 1$ are anti-parallel above w on a run starting at BL with $w + 1$ interior, and (ALT) again gives $r_{w+1}(u + \frac{1}{2}) = -\beta \neq \tau$. The case TL a c -vertex is the reflection. If both BR and TL are c , the c -relations give $\tau = -\beta$ and $\rho = \beta$, and Q is good. $\square$

So on a thin cell, with a single kink, frozen is the same as even curl, the a - and b -vertices lie on complete diagonals of the two sublattices separately, and the state is determined up to the global arrow reversal by its two sets of diagonals: on $(1, 2, 2)$ the 698 frozen states with a kink fall into 349 such classes of two, and on $(1, 3, 3)$ the 12,026 into 6,013. The proof is global at one point, the induction that runs around the torus, and the solver agrees that it must be: a bad square forces none of its eight neighbouring squares bad.

6 Theorem H

Theorem H (certified on five cells). A frozen state in which all six directions occur has even curl on every hexagon. Equivalently, in such a state the types of the pass-through vertices along every chain change only across a kink of that chain.

The negation, “frozen, all six directions present, some hexagon odd”, is unsatisfiable with DRAT proofs checked by drat-trim on $(1, 2, 2)$, $(1, 3, 3)$, $(2, 2, 2)$, $(2, 2, 3)$ and $(2, 2, 4)$, and unsatisfiable without proof logging on $(2, 3, 3)$ under the hypothesis that every chain carries a kink. On $(3, 3, 3)$ and $(4, 4, 4)$ the solver gave no verdict in four and ten hours. Theorem 2D is the case $a = 1$, where a single kink replaces the six directions.

The hypothesis is sharp in a strong sense. A single distance-two type change on a chain is compatible with any five of the six directions $((2, 2, 2), (2, 2, 4))$; pairs and triples also on $(3, 3, 3)$, and only the full set of six is excluded; the per-axis hypotheses “both signs of x ” and “every x -chain kinked” do not suffice; and a single odd hexagon forces no particular chain kink-free. So a proof of H cannot exclude a direction locally, and, like Lemma L, must

be an argument over the whole torus. One local fact is available: in a frozen state an odd hexagon always has an odd hexagon among the six that share two consecutive bonds with it, the shared pair sitting at one of its four pass-through vertices (solver-established with freezing on the 37 hexagons around it only, on $(3, 3, 3)$ and $(4, 4, 4)$), so the odd hexagons form closed trails on the torus; the proof would show that such a trail forces a direction to be absent. The two-dimensional proof is its model: the facts (AP), (END) and (ALT) hold for any pair of adjacent parallel chains, with the difference that in three dimensions a chain has rungs to a given partner only at every second vertex, the other vertices' rungs going to a partner in the next sheet, so (END) applies only to kinks at rung positions.

7 Theorem P and the slab sums

Theorem P (certified on six cells). An ice state with even curl on every hexagon, in any flux sector and with no freezing hypothesis, on a cell whose extents are all at least two, is invariant under a one-cube coordinate shift $(4, 0, 0)$, $(0, 4, 0)$ or $(0, 0, 4)$.

The negation is unsatisfiable with DRAT proofs checked by drat-trim on $(2, 2, 2)$, $(2, 2, 3)$, $(2, 2, 4)$, $(2, 3, 3)$, $(3, 3, 3)$ and $(4, 4, 4)$ (the last in 959 s of CaDiCaL), and unsatisfiable on $(2, 4, 4)$ and $(3, 3, 4)$. On thin cells the negation is satisfiable, as it must be: the aperiodic frozen states of $(1, 3, 3)$ have even curl by Theorem 2D.

The explanation is linear algebra over $\mathbb{F}_2$. Let K be the space of $s \in \mathbb{F}_2^E$ with $\sum_{e \ni v} s_e = 0$ at every vertex and $\kappa_h(s) = 0$ on every hexagon. A slab meets every vertex in zero or two bonds and every hexagon in zero or two, so the $4a + 4b + 4c$ slab indicators lie in K . Gaussian elimination gives:

cell	dim K	slab rank	slabs span K
$(1,2,2)$	16	16	yes
$(2,2,2)$	26	20	no
$(2,2,3)$	32	24	no
$(2,2,4)$	42	28	no
$(2,3,3)$	28	28	yes
$(3,3,3)$	32	32	yes

The vectors of K outside the slab span appear only with two even extents and are ribbons, a single line per slab restricted to a drifting band of four slabs. They carry no ice state: every even-curl ice state enumerated is a slab sum (all 714 of $(1, 2, 2)$, all 1,266 of $(2, 2, 2)$), and the formula “ice, even curl, some linear functional vanishing on the slab span equal to one” is unsatisfiable on $(2, 2, 2)$, $(2, 2, 3)$, $(2, 2, 4)$, $(2, 4, 4)$ and $(4, 4, 4)$. So on every cell tested:

P1 (checked, not proved). Every even-curl ice state is a slab sum, $s_e = X_{x(e)} + Y_{y(e)} + Z_{z(e)} \pmod{2}$, for three profiles $X : \mathbb{Z}_{4a} \rightarrow \mathbb{F}_2$, Y, Z on the layers, $x(e)$ being the x -slab of e .

For a slab sum the ice rule is a condition on the jumps $\delta X_x = X_x + X_{x-1}$. At the A vertex (x, y, z) the four bits are $a+b+c, a+b'+c', a'+b+c', a'+b'+c$ with $a = X_x$, $a' = X_{x-1}$ and likewise for b, c ; they are all equal iff $\delta X_x = \delta Y_y = \delta Z_z$, and the same holds at B vertices. Since their sum is always even, the ice rule says exactly:

(ICE) $(\delta X_x, \delta Y_y, \delta Z_z)$ is not constant at any vertex, the vertices being the even triples with $x + y + z \equiv 0$ and the odd triples with $x + y + z \equiv 3 \pmod{4}$.

The type of a vertex is the axis whose jump is the odd one out, and a one-cube period along x is $X_{x+4} = X_x$.

Profile lemma. Every slab sum satisfying (ICE) has a 4-periodic profile.

Proof. Let $V_X(r) = \{\delta X_x : x \equiv r \pmod{4}\}$, a nonempty subset of $\{0, 1\}$, and likewise V_Y, V_Z . (ICE) says the triple intersection $V_X(r_x) \cap V_Y(r_y) \cap V_Z(r_z)$ is empty for the four even residue triples $(0, 0, 0), (0, 2, 2), (2, 0, 2), (2, 2, 0)$ and the four odd ones $(1, 1, 1), (1, 3, 3), (3, 1, 3), (3, 3, 1)$. The profile X is 4-periodic iff all $V_X(r)$ are singletons $\{e_r\}$ with $e_0 + e_1 + e_2 + e_3 = 0$.

If $V_X(0) = \{0, 1\}$, the triples $(0, 0, 0)$ and $(0, 2, 2)$ force $V_Y(0) = \{v\}$, $V_Z(0) = \{1 - v\}$, $V_Y(2) = \{w\}$, $V_Z(2) = \{1 - w\}$, and $(2, 0, 2), (2, 2, 0)$ force $v = w$, since $w = 1 - v$ would require both $v \notin V_X(2)$ and $1 - v \notin V_X(2)$. So a doubleton on one axis makes the other two axes constant on that parity class, with opposite values; the odd class has the same structure.

Case 1, a doubleton exists, say $V_X(0) = \{0, 1\}$, so $Y \equiv v$ and $Z \equiv 1 - v$ on even coordinates. If some axis has an odd doubleton, the other two are constant on odd coordinates, and one of Y, Z is then constant on each parity class with values (v, v', v, v') ,

sum zero, hence periodic. If there is no odd doubleton, Y has odd values (p, p') and Z has (q, q') ; if $p = p'$ or $q = q'$ we are done; otherwise $p' = 1 - p$, $q' = 1 - q$, and the odd conditions read: not $e_X(1) = p = q$; not $e_X(1) = 1 - p = 1 - q$; not $e_X(3) = p = 1 - q$; not $e_X(3) = 1 - p = q$. If $p = q$ the first two exclude both values of $e_X(1)$; if $p \neq q$ the last two exclude both values of $e_X(3)$. A doubleton on an odd class is the mirror case.

Case 2, all twelve sets are singletons. For the even class write $\sigma_\alpha = e_\alpha(0) + e_\alpha(2)$. If all three $\sigma_\alpha = 1$, with $(p, q, r) = (e_X(0), e_Y(0), e_Z(0))$ the four even conditions are: not $p = q = r$; not $q = r = 1 - p$; not $p = r = 1 - q$; not $p = q = 1 - r$; and every triple in $\mathbb{F}_2^3$ violates one of them. If exactly two axes, say X, Y , have $\sigma = 1$ and $Z \equiv z$, the conditions $(0, 0, 0)$ and $(0, 2, 2)$ force $e_X(0) = 1 - z$, then $(2, 0, 2)$ forces $e_Y(0) = 1 - z$, and $(2, 2, 0)$ reads not $z = z = z$. So at most one axis has $\sigma_\alpha = 1$, and likewise at most one has $\sigma'_\alpha = e_\alpha(1) + e_\alpha(3) = 1$. If no axis were periodic, $\sigma'_\alpha = 1 - \sigma_\alpha$ for all three, giving $\sum_\alpha \sigma'_\alpha = 3 - \sum_\alpha \sigma_\alpha \geq 2$. So some axis has $\sigma_\alpha = \sigma'_\alpha$, that is, $e_0 + e_1 + e_2 + e_3 = 0$. $\square$

Brute force over all jump vectors confirms the lemma on $(1, 2, 2)$, $(1, 3, 3)$, $(2, 2, 2)$, $(2, 2, 3)$ and $(2, 3, 3)$ (2,532 slab-sum ice states on $(2, 2, 2)$ up to the three global constants, 27,492 on $(2, 3, 3)$, none without a 4-periodic profile). On an odd extent the parity of $\sum_x \delta X_x$ makes all-singleton value sets periodic at once, which is why odd cells carry no anti-periodic states, while the even cells do: $X_{x+4} = X_x + 1$ is the case $\sum_r e_r = 1$ of Case 2.

Theorem P is therefore proved for slab sums, and P1 is what stands between the certificates and a proof for all cells.

8 Consequences

Theorems H and P together give: a frozen state in which all six directions occur has a one-cube coordinate period. With Lemma A and the remark on zero flux, every frozen zero-flux state of a cell whose extents are all at least two is invariant under a nonzero translation, either the half-box diagonal shift or a one-cube shift. That is the periodicity conjecture, and more: such a state is either the lift of a state of a cell one cube thick, where Theorem 2D classifies it, or a state of Lemma A. On the evidence of P1 the frozen states with six directions are exactly the slab sums, three one-dimensional profiles whose jumps are never all equal at a vertex.

At the cubic cell of side four, where a cube-and-conquer proof of the conjecture left 407 of 23,153 leaf cubes unrefuted, Theorem P is certified and the conjecture depends on the single certificate of Theorem H at (4, 4, 4).

9 Related work

The Coulomb phase of spin ice and its emergent gauge field are reviewed by Henley (2010) and Castelnovo, Moessner, and Sondhi (2012); the hexagon-flip chain is the classical face of the Rokhsar–Kivelson point of quantum spin ice (Rokhsar and Kivelson 1988; Henley, Fisher, and Balents 2004; Henley 2004), and loop and hexagon updates are the standard ergodic moves for ice models (Barkema and Newman 1998). The [100] projection used here, in which the diamond lattice is a stack of square lattices whose rows and columns alternate between layers, is the geometry of the Kasteleyn transition of spin ice in a [100] field (Jaubert et al. 2008), where the same $\langle 110 \rangle$ chains are the strings that carry the transition; the kagome layers of the [111] projection, in which the hexagons lie, are the geometry of the [111] plateau (Moessner and Sondhi 2003; Fennell et al. 2007). Square ice with its height function is the six-vertex model, whose height representation and its delocalisation are treated by Duminil-Copin et al. (2024); the domino moves of the thin cell, as opposed to plaquette moves, are the moves of domino tilings in the sense of Elkies et al. (1992) and Saldanha et al. (1995) rather than of square ice, and the classification of Theorem 2D by two families of diagonals has the flavour of the frozen regions of those tilings. Parity constraints on plaquettes of the kind the curl imposes are the elementary objects of $\mathbb{Z}_2$ gauge theory (Wegner 1971) and of Kasteleyn’s treatment of dimer statistics (Kasteleyn 1963); the counting argument of Lemma L, a global average forcing a local equality, is of the kind used to show that local rules on a torus have no room to vary. Frozen configurations under local moves are studied for triangulations by Kownacki (2004); for spin ice the dynamics that freezes at low temperature is the monopole dynamics of Jaubert and Holdsworth (2009) and Melko and Gingras (2004), which is a different mechanism, since here the ice rule is never broken and freezing is a property of the hexagon move itself.

10 Certification

Every claim marked certified rests on a CNF formula built by the repository code, an UNSAT verdict from CaDiCaL with a DRAT proof, and a check of that proof by drat-trim, which accepts a proof only if every clause it adds follows from the formula and the clauses before it by reverse unit propagation or is a resolution asymmetric tautology, and the empty clause is derived. What is trusted is the formula's construction and the checker. The formulas are: for P, the ice rule (exactly two of four bits set at every vertex), one parity constraint per hexagon (even), and, for each one-cube shift, one auxiliary per bond witnessing a disagreement between the bond and its translate, with the clause that for every shift some witness is true; for H, the ice rule, two six-clauses per hexagon forbidding the two circulating patterns, one selector per hexagon implying odd parity with the clause that some selector is true, and for each of the six directions one selector per vertex implying that direction there with the clause that some vertex carries it. The formulas, the drat-trim verdict files and the SHA-256 of each proof are in the repository; the proofs are regenerated by `experiments/08_frozen_structure/certify_curl.py`, and the same statements without proof logging, together with the $\mathbb{F}_2$ computations and the profile check, by the stages `curl`, `slabspace` and `profiles` of `run_structure.py`.

The proofs in this note (Lemmas A, C and L, Theorem 2D, the chain form and the profile lemma) are hand proofs; their finite instances were also checked by solver as stated.

11 Artefacts

- `docs/frozen-structure-2026-09-05.md`: the working note, every claim with its status.
- `icemix/cnf.py`: the encoding; `experiments/08_frozen_structure/structure_queries.py`: the `curl`, `slab` and `profile` helpers; `run_structure.py`: the stages; `certify_curl.py`: the DRAT certificates.
- `experiments/08_frozen_structure/results/curlP_*`, `curlH_*`, `curl_certificates.json`, `drat_sha256.txt`, `curl_*.json`, `slabspace.json`, `profiles.json`.

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